TLE / ICT 9
FOURTH QUARTER

LESSON 1

JAVASCRIPT
Introduction

Purpose

To create dynamic, interactive user experiences, handling everything from UI rendering (like React for Facebook) and content loading to real-time updates, personalized feeds, video controls (captions, volume), and tracking user activity (Meta Pixel).

Examples

Static Website
HTML
JScript
CSS
VS
HTML
Python
MySQL
CSS
C++
Dynamic Website

A static web page, sometimes called a flat page or a stationary page, is a web page that is delivered to a web browser exactly as stored.
Example:
Photographer's portfolio

VS

A dynamic website is a type of website that displays different content and interacts with users based on their actions, preferences, or other variables.
Example:
Social Media Platforms

JavaScript

- The MOST popular & widely used scripting language on the internet.
- Recognized & works in all major browsers.

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JavaScript

- Developed by Brendan Eich of Netscape under the name Mocha, which was later renamed to LiveScript.

JavaScript

- Opened to Internet Explorer in 1996.
- Enhances webpages and can be used to build video games.

JavaScript

- A trademark of the Oracle Corporation and is licensed for use to other entities like Mozilla Foundation.

JavaScript

- A client-side, high-level scripting, interpreted & object-oriented.

Client-side – the codes and formulas are processed right on the user's own computer.

JavaScript

High-level scripting – codes are written in words that are close to English as possible.

Interpreted – the program is passed as source code then converted into machine code.

JavaScript

- Designed to add interactivity and functionality to the Web site. Thus making the contents more dynamic and the elements are enhanced.

JavaScript

- Case-sensitive.
- Semi-colons at the end if each statement are not required.

What can JavaScript do?

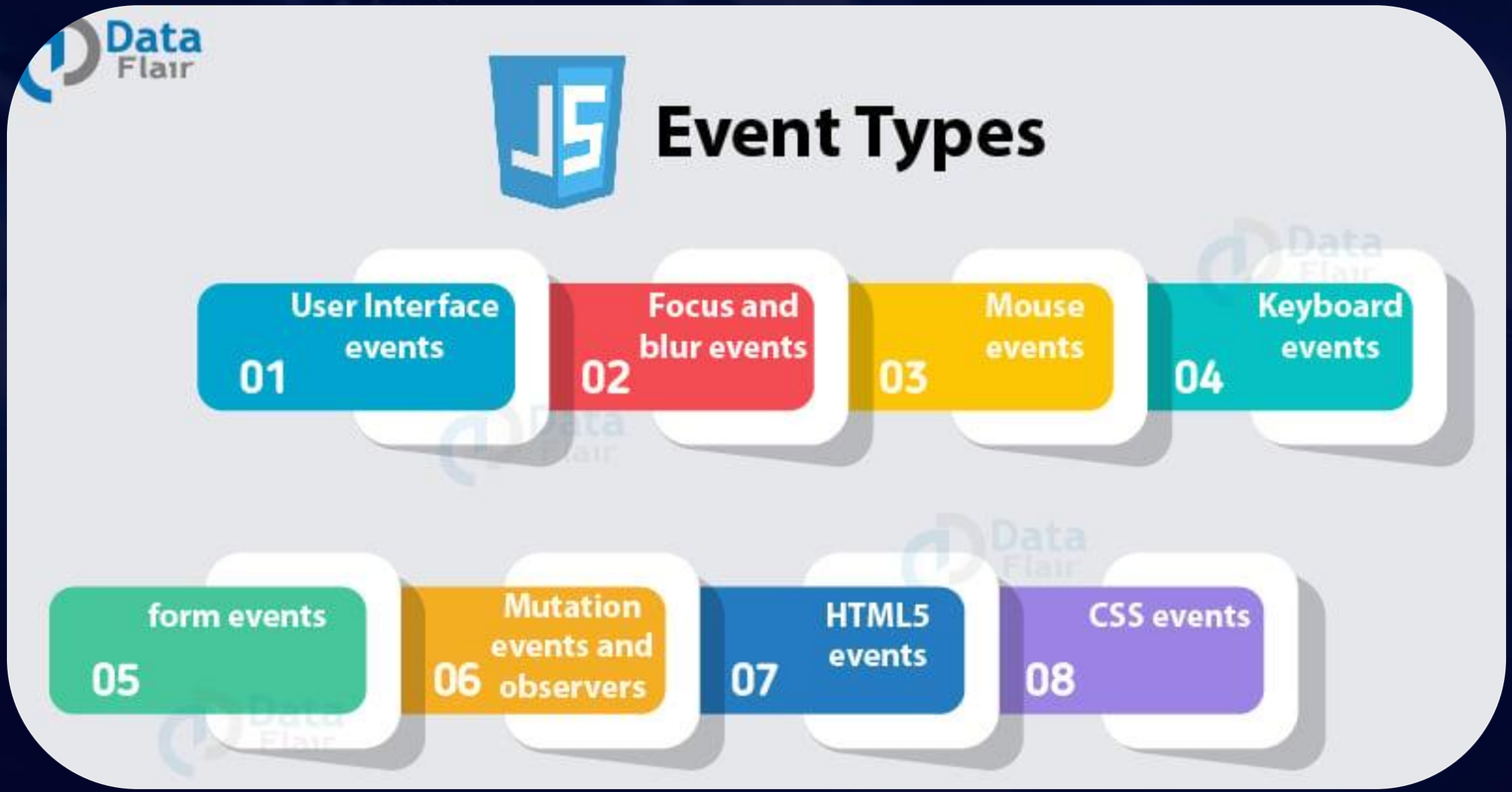
- Can react to events – certain events such as mouse clicks, and pre-loading of webpage can lead to executing codes written in JavaScript.

What can JavaScript do?

- Can react to events
- Can be used to validate data
- Can be used to create cookies
- Can enhance a webpage.

Events

These are mouse clicks, and pre-loading of webpage that can lead to executing codes within JavaScript.



Form Validation

A form requires user input, instead of using server-side scripting, JavaScript can be used to validate it and thus save processing time.

Cookies

Used to save or retrieve information from a visitor's computer. This can be used to monitor the visitor's internet habit and preferences.

Cookies

Also known as 'internet cookies', cookies are text files placed on devices (eg computers or smartphones) by websites. They hold data related to a user's interactions with the website that placed the cookies.

- : JavaScript is used extensively in the Google search engine for real-time updates and efficient query handling.
- : This popular video platform uses JavaScript for seamless, uninterrupted music and video streaming, dynamic content, and drag-and-drop functionality for playlists.
- : Social media platforms rely on JavaScript for features like real-time content updates, infinite scrolling, and interactive menus. Facebook even developed its own JavaScript framework called React.

Features of JavaScript

- Supports all structured programming syntax in C
- Has dynamic typing, the value of the variable is dependent on what the value was assigned to it even during run time, the type can still change.

Features of JavaScript

- Can run locally in a Web site, so interaction within the user and the site is faster or more responsive.
- Can detect user actions, such as mouse clicks, which HTML could not do alone.

Features of JavaScript

- Can be combined with CSS to produce DHTML. DHTML can make site more flexible by adding effects such as message scrollers, mouse trails and falling snow.

Characteristics of JavaScript

- Lightweight programming language
- Inserted into HTML pages
- Supported by all modern web browsers
- Easy to learn and encoder-friendly

Characteristics of JavaScript

- Quick to download effects
- No need for any extra tools to write
- License-free
- Has a thriving and supportive online community of JavaScript developers and information resources

Disadvantages of JavaScript

- Some browsers do not support JS. Examples are PDA and mobile phones do not execute JS. Some browsers disable execution of JS as a security precaution.
- Any secret embedded in JS could be extracted by a determined adversary

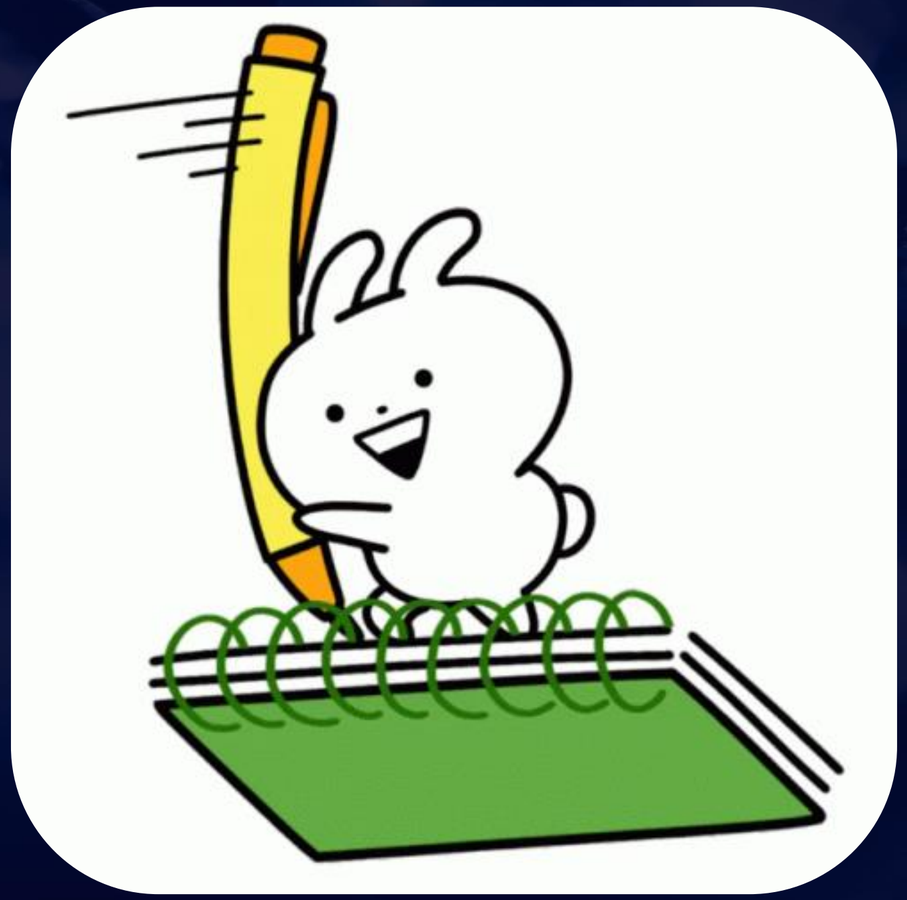
Disadvantages of JavaScript

- JS and DOM (Document Object Model) provide potential for malicious authors to deliver scripts to run on a client computer via web.
- Web site authors cannot perfectly conceal how their JS operates because the code is sent to the client.

Disadvantages of JavaScript

- Source codes that have been deliberately made hard to understand can be reverse engineered so your codes are still exposed to possible threat.

Note



A script is a set of computer instructions or commands that puts together or connects existing components to accomplish a new related task. JavaScript is a powerful scripting language used to create web functionalities. A text editor and a web browser are the tools needed in JavaScripting. The basic components of JavaScript language are objects, methods and properties.

How to start using JavaScript

<script></script> is used to connote JavaScript. We use the *type* attribute to define the scripting language and identify its version. To display an output in a web page, the *document.write* command should be used.

JavaScript.txt

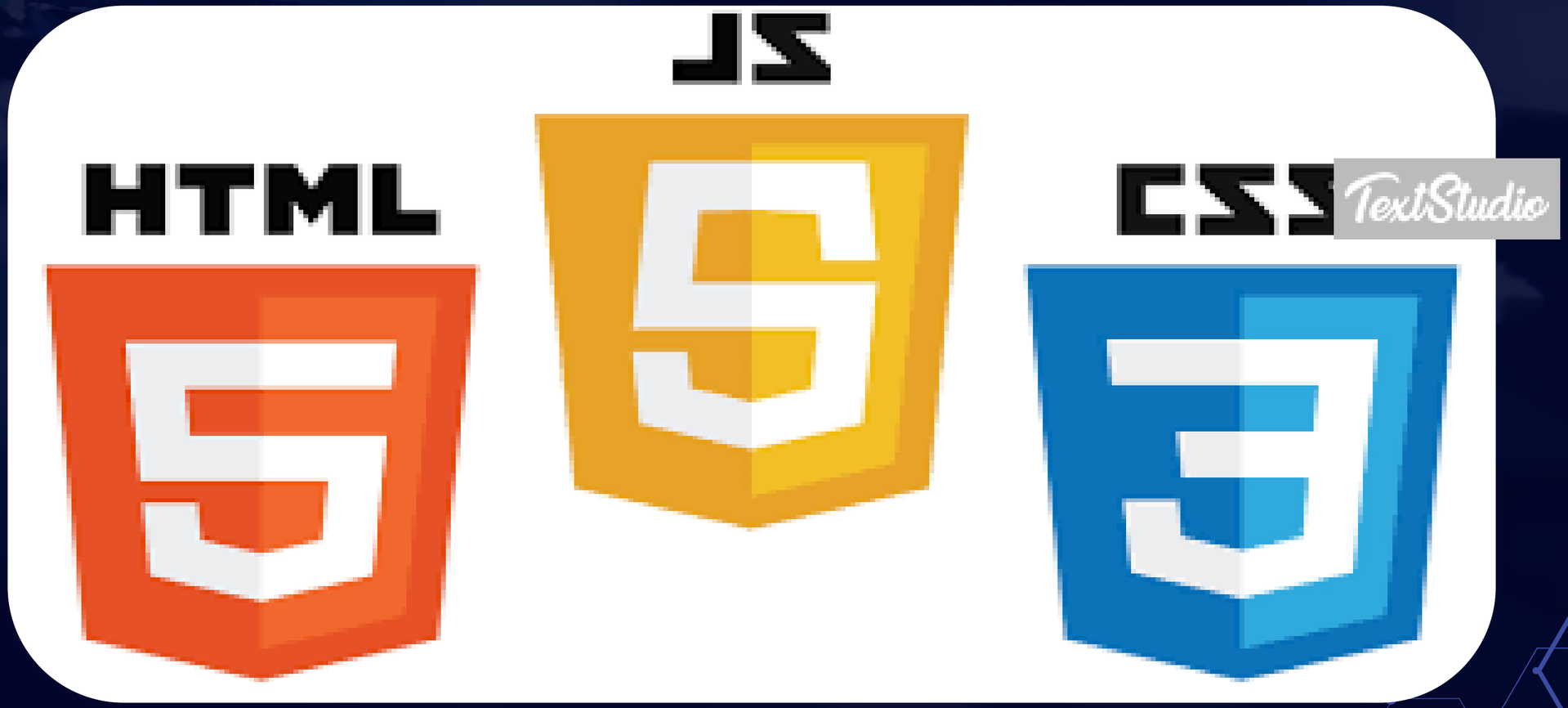
```
<!DOCTYPE html>
<html>
<head><title>JavaScript</title>
</head>
<body>
<h1>My First JavaScript code</h1>
<script type="text/JavaScript">
<!-
document.write("Displaying text using
                JavaScript");
//->
</script>
</body>
</html>
```

JavaScript

(1) Facebook

My First JavaScript code

Difference



What's the Difference?

HTML
Hypertext Markup Language

Create the structure

- Controls the layout of the content
- Provides structure for the web page design
- The fundamental building block of any web page

CSS
Cascading Style Sheet

Stylize the website

- Applies style to the web page elements
- Targets various screen sizes to make web pages responsive
- Primarily handles the "look and feel" of a web page

Javascript

Increase interactivity

- Adds interactivity to a web page
- Handles complex functions and features
- Programmatic code which enhances functionality